Puppet Classification Banner Module

RHEL 8 / 9 / 10 - custom OpenVox/Puppet module documentation

Purpose: Install and configure the GNOME classification banner for login and local user sessions.

Module layout

```
custom/
  manifests/classification_banner.pp
  manifests/dconf.pp
  templates/classification-banner.epp
```

Configuration

```
include custom::classification_banner

# Override profiles through Hiera when site-specific text or colors are
needed.
custom::classification_banner::manage_gui: true
```

Behavior and safety: A trusted pp_network value selects one of four typed banner profiles. Puppet installs the extension, manages both GDM and local dconf databases and locks, and runs dconf update when content changes. RHEL 8 requires release 8.6 or newer.

Supported platform: Red Hat Enterprise Linux 8, 9, and 10. Generated from the repository sources; rerun **tools/generate_custom_module_docs.ps1** after code changes.

Classification Banner

custom/manifests/classification_banner.pp

```
# @summary
#   Installs and configures GNOME classification banners by trusted network.
#
# @param manage_gui
#   Whether to manage the graphical banner. Defaults to the custom_gui fact.
# @param pp_network
#   Network profile name. Defaults to the pp_network trusted extension.
# @param network_profiles
#   Banner settings keyed by the pp_network trusted extension.
class custom::classification_banner (
  Boolean $manage_gui = $facts['custom_gui'] ? {
    true    => true,
    default => false,
  },
  Optional[String] $pp_network = $trusted['extensions']['pp_network'],
  Hash[Enum['network1', 'network2', 'network3', 'network4'], Struct[{
    'background_color' => String[1],
    'message'          => String[1],
    'top_banner'       => Boolean,
    'bottom_banner'    => Boolean,
    'system_info'      => Boolean,
    'color'            => String[1],
  }]] $network_profiles = {
    'network1' => {
      'background_color' => 'rgba(200,16,46,0.75)',
      'message'          => 'Network 1 Banner message',
      'top_banner'       => true,
      'bottom_banner'    => false,
      'system_info'      => true,
      'color'            => 'rgb(255,255,255)',
    },
    'network2' => {
      'background_color' => 'rgba(200,16,46,0.75)',
      'message'          => 'Network 2 Banner message',
      'top_banner'       => true,
      'bottom_banner'    => false,
      'system_info'      => true,
      'color'            => 'rgb(255,255,255)',
    },
    'network3' => {
      'background_color' => 'rgba(200,16,46,0.75)',
      'message'          => 'Network 3 Banner message',
      'top_banner'       => true,
      'bottom_banner'    => false,
      'system_info'      => true,
      'color'            => 'rgb(255,255,255)',
    },
    'network4' => {
      'background_color' => 'rgba(200,16,46,0.75)',
      'message'          => 'Network 4 Banner message',
      'top_banner'       => true,
      'bottom_banner'    => false,
      'system_info'      => true,
      'color'            => 'rgb(255,255,255)',
    },
  },
) {
  if $facts['os']['family'] != 'RedHat' {
    fail('custom::classification_banner supports only Red Hat family operating systems')
  }

  $os_major = $facts['os']['release']['major']
```

```
case $os_major {
  '8': {
    if Integer($facts['os']['release']['minor']) < 6 and $manage_gui {
      fail('custom::classification_banner requires RHEL 8.6 or newer because the package is unavailable on earlier RHEL 8 releases')
    }
  }
  '9', '10': { }
  default: {
    fail('custom::classification_banner supports only RHEL major versions 8, 9, and 10')
  }
}

if $manage_gui {
  if $pp_network == undef or $pp_network == '' {
    fail('custom::classification_banner requires the pp_network trusted extension')
  }
  if !($pp_network in $network_profiles) {
    fail("custom::classification_banner has no banner profile for pp_network '${pp_network}'")
  }

  $profile = $network_profiles[$pp_network]
  $banner_content = epp('custom/classification-banner.epp', $profile)
  $lock_content = @("LOCKS"/L)
    # Managed by CES Puppet.
    # Manual changes will be replaced.
    /org/gnome/shell/enabled-extensions
    /org/gnome/shell/extensions/classification-banner/background-color
    /org/gnome/shell/extensions/classification-banner/message
    /org/gnome/shell/extensions/classification-banner/top-banner
    /org/gnome/shell/extensions/classification-banner/bottom-banner
    /org/gnome/shell/extensions/classification-banner/system-info
    /org/gnome/shell/extensions/classification-banner/color
    | LOCKS

  contain custom::dconf

  package { 'gnome-shell-extension-classification-banner':
    ensure => installed,
    notify => Exec['update custom dconf databases'],
  }

  file { 'custom gdm classification banner':
    ensure  => file,
    path    => '/etc/dconf/db/gdm.d/99-class-banner',
    owner   => 'root',
    group   => 'root',
    mode    => '0644',
    content => $banner_content,
    require => Package['gnome-shell-extension-classification-banner'],
    notify  => Exec['update custom dconf databases'],
  }

  file { 'custom gdm classification banner lock':
    ensure  => file,
    path    => '/etc/dconf/db/gdm.d/locks/99-class-banner',
    owner   => 'root',
    group   => 'root',
    mode    => '0644',
    content => $lock_content,
    require => Package['gnome-shell-extension-classification-banner'],
    notify  => Exec['update custom dconf databases'],
  }

  file { 'custom local classification banner':
    ensure  => file,
    path    => '/etc/dconf/db/local.d/99-class-banner',
```

```
      owner   => 'root',
      group   => 'root',
      mode    => '0644',
      content => $banner_content,
      require => Package['gnome-shell-extension-classification-banner'],
      notify  => Exec['update custom dconf databases'],
    }

    file { 'custom local classification banner lock':
      ensure  => file,
      path    => '/etc/dconf/db/local.d/locks/99-class-banner',
      owner   => 'root',
      group   => 'root',
      mode    => '0644',
      content => $lock_content,
      require => Package['gnome-shell-extension-classification-banner'],
      notify  => Exec['update custom dconf databases'],
    }
  }
}
```

```
# @summary
#   Creates the shared dconf directories and refresh command used by this module.
class custom::dconf {
  file { 'custom dconf directory':
    ensure => directory,
    path   => '/etc/dconf',
  }

  file { 'custom dconf database directory':
    ensure => directory,
    path   => '/etc/dconf/db',
  }

  file { 'custom gdm dconf directory':
    ensure => directory,
    path   => '/etc/dconf/db/gdm.d',
  }

  file { 'custom gdm dconf locks directory':
    ensure => directory,
    path   => '/etc/dconf/db/gdm.d/locks',
  }

  file { 'custom local dconf directory':
    ensure => directory,
    path   => '/etc/dconf/db/local.d',
  }

  file { 'custom local dconf locks directory':
    ensure => directory,
    path   => '/etc/dconf/db/local.d/locks',
  }

  exec { 'update custom dconf databases':
    command     => '/usr/bin/dconf update',
    refreshonly => true,
    require     => [
      File['custom gdm dconf directory'],
      File['custom gdm dconf locks directory'],
      File['custom local dconf directory'],
      File['custom local dconf locks directory'],
    ],
  }
}
```

Classification Banner

custom/templates/classification-banner.epp

```
<%- |
  String[1] $background_color,
  String[1] $message,
  Boolean   $top_banner,
  Boolean   $bottom_banner,
  Boolean   $system_info,
  String[1] $color,
| -%>
# Managed by CES Puppet.
# Manual changes will be replaced.
[org/gnome/shell]
enabled-extensions=['classification-banner@gnome-shell-extensions.gcampax.github.com']

[org/gnome/shell/extensions/classification-banner]
background-color='<%= $background_color %>'
message='<%= $message %>'
top-banner=<%= $top_banner %>
bottom-banner=<%= $bottom_banner %>
system-info=<%= $system_info %>
color='<%= $color %>'
```